

Particle in free space

1. Separable Solution – Travelling wave
2. General solution – Wavepacket
3. Gaussian wavepacket
4. Heisenberg uncertainty principle

$$g = \tilde{D}(f) = \frac{d}{dx}(x^4) = 4x^3$$

$$\frac{d}{dx}(\sin^3 x) = 3\sin^2 x \cos x$$

$$\frac{d}{dx}(e^{-x^2})$$

$$= \frac{d}{dx} f(g(x))$$

$$\tilde{I}(f) = \int f(x) dx = g(x)$$

$$g(x)f(x) = \frac{d}{dx} g(x)^2 = \tilde{D}(g)$$

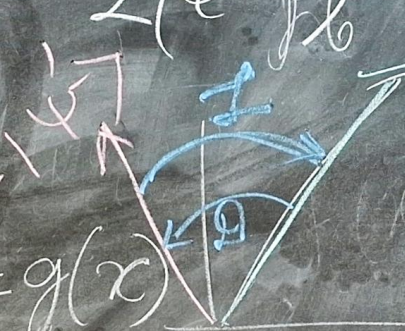
$$\frac{d}{dx} f(x) \bigg|_{g(x)} \cdot \frac{dg(x)}{dx} = -e^{-x^2} \cdot 2x = -2xe^{-x^2}$$

$$\frac{d}{dx}(f \circ g) ; f(x) = x^3, g(x) = \sin x$$

$$\mathcal{L}(x^4) = \frac{d}{dx} f(g(x)) = \frac{d f(x)}{dx} \bigg|_{g(x)} \cdot \frac{dg}{dx} \bigg|_x \leftarrow \text{Chain rule}$$

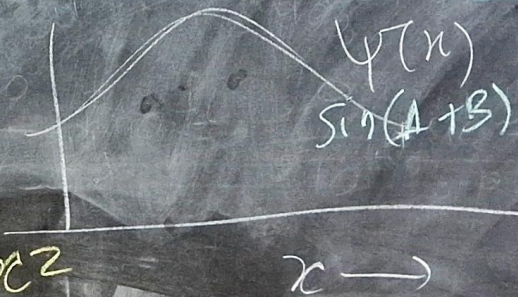
$$\mathcal{L}(\sin^3 x)$$

$$\mathcal{L}(e^{-x^2}) = \text{function space}$$



$$|\psi\rangle = |f\rangle$$

$$\langle x | \psi \rangle =: \psi(x)$$



$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin 3x = \sin(x+2x)$$

$$\sin 2x = 2\sin x \cos x$$

$$\sin 3x = \sin(2x+x)$$

$$= \sin 2x \cos x + \cos 2x \cdot \sin x$$

$$= 2\sin x \cos^2 x + (1-2\sin^2 x) \cdot \sin x$$

$$= 2\sin x(1-\sin^2 x) + \sin x - 2\sin^3 x.$$

$$\sin 3x = 3\sin x - 4\sin^3 x$$

$$\int \sin^3 x dx = \int \left(\frac{3\sin x - \sin 3x}{4} \right) dx = -\frac{3}{4} \cos x + \frac{1}{12} \cos 3x + C$$

$$\int \cos \alpha x dx = \frac{\sin \alpha x}{\alpha}$$

$$\int \sin \alpha x dx = -\frac{\cos \alpha x}{\alpha}$$

$$\int \sin 2x dx = -\frac{\cos 2x}{2}$$

Quantum Structures.

— To solve

$$H \psi(x) = E \psi(x)$$

$$\text{i.e. } \left[\frac{-\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \psi(x) = E \psi(x),$$

we need $V(x)$, the potential energy function.

This $V(x)$ defines the quantum structure that the particle resides in.

The simplest structure is by logic, "no structure".

$$\text{i.e. } V(x) = 0$$

i.e. particle is in free space

Particle in free space:

If $V(x) = 0$,

TISE reduces to

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E \psi(x).$$

$$\frac{d^2}{dx^2} \psi(x) = -k^2 \psi(x), \text{ where}$$

$$k^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow E = \frac{\hbar^2 k^2}{2m}$$

Therefore, a general (g)
solution will be given

by

$$\psi_g(x) = A e^{+ikx} + B e^{-ikx}$$

$$\text{and } E_g = \frac{\hbar^2 k^2}{2m}$$

This differential equation has

particular (p) solutions

where $k \in \mathbb{R}$

$$\psi_{p_1}(x) = e^{+ikx}$$

$$\psi_{p_2}(x) = e^{-ikx}$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\Rightarrow \cos\theta = \frac{e^{+i\theta} + e^{-i\theta}}{2}$$

$$\sin\theta = \frac{e^{+i\theta} - e^{-i\theta}}{2i}$$

why?

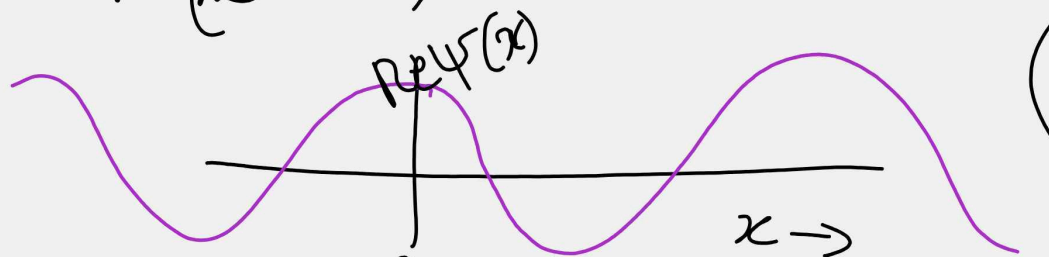
particular

solutions are $\sin kx, \cos kx$

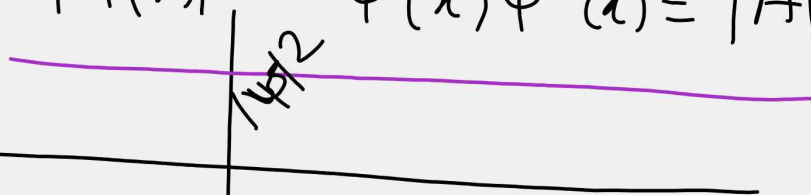
Since

$$\frac{d}{dx} \left(\frac{d}{dx} \sin kx \right) = \frac{dk \cos kx}{dx} = -k^2 \sin kx$$

$$\text{Re}(A e^{i k x}) = A \cos kx$$



$$|\psi(x)|^2 = \psi(x) \psi^*(x) = |A|^2$$



But

$$\langle \psi | \psi \rangle = \int_{-\infty}^{+\infty} |\psi|^2 dx = \int_{-\infty}^{+\infty} |A|^2 dx = |A|^2 (\infty)$$

Implications

- Particular solutions cannot represent physically acceptable states
- However, we can create a L.C. of different k states so that the norm is 1.

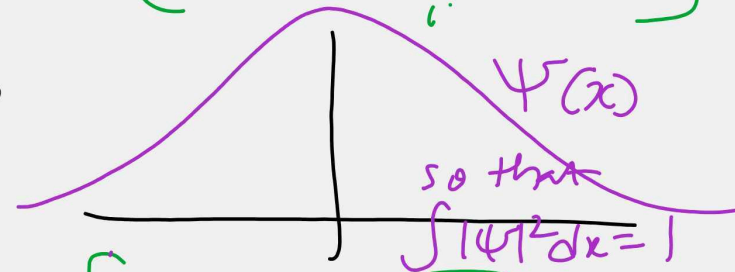
Wavepacket

⊛ (ii) Integrate over all the x gives
$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi(x) e^{-ik_0 x} dx = \phi(k_0)$$

— A wavepacket is a combination of stationary states of different k 's.

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{ikx} dx$$

Recall
$$|\psi\rangle = \sum_i C_i |i\rangle$$



— How to determine the coefficients $\phi(k)$?

(i) Left Multiply on both sides by $e^{-ik_0 x}$

$$\begin{aligned} \psi(x) e^{-ik_0 x} dx &= \frac{1}{\sqrt{2\pi}} \int \phi(k) e^{ikx} e^{-ik_0 x} dx \\ &= \frac{1}{\sqrt{2\pi}} \int \phi(k) e^{i(k-k_0)x} dx = \frac{1}{\sqrt{2\pi}} \phi(k) \delta(k-k_0) \end{aligned}$$

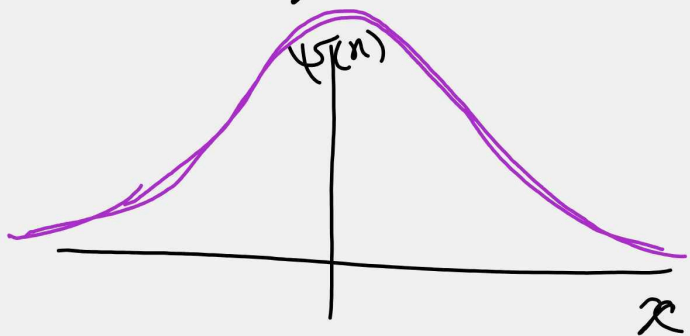
$$\langle j | \psi \rangle = \sum_i C_i \langle j | i \rangle$$

$\star$

Gaussian wave packet.

— Gauss

$$\psi(x) = A e^{-ax^2}, \quad a \in \mathbb{R}, A \in \mathbb{R}, a > 0$$



As $x \rightarrow \infty$, $\psi(x) \rightarrow 0$

As $x \rightarrow -\infty$, $\psi(x) \rightarrow 0$

At $x=0$, $\psi(x) = A$

Gaussian integral

$$\int_0^{\infty} x^{2n} e^{-\frac{x^2}{2a}} dx = \sqrt{\pi} \frac{(2n)!}{n!} \left(\frac{a}{2}\right)^{n+\frac{1}{2}}$$

so $n=0$,

$$\int_0^{\infty} e^{-\frac{x^2}{2a}} dx = \sqrt{\pi} \frac{2}{1} \left(\frac{a}{2}\right)^{\frac{1}{2}}$$

(i) Normalization const.

$$\int |\psi|^2 dx = 1$$

$$\int_{-\infty}^{\infty} A^2 e^{-2ax^2} dx = 1$$

$$A^2 \int_{-\infty}^{\infty} e^{-2ax^2} dx = 1 \Leftrightarrow 2A^2 \int_0^{\infty} e^{-\frac{x^2}{b^2}} dx = 1,$$

$$\frac{2A^2 \sqrt{\pi}}{\sqrt{2a}} = 1 \Leftrightarrow A = \left(\frac{a}{2\pi}\right)^{\frac{1}{4}}$$

where $b^2 = \frac{1}{2a}$, so that

Heisenberg uncertainty principle. for Gaussian wavepacket

$$\psi(x) = \left(\frac{a}{2\pi}\right)^{1/4} e^{-ax^2}$$

Claim: $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$,

where

$$\Delta x = \sqrt{\langle (x - \langle x \rangle)^2 \rangle} = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

|| only $\Delta p = \sqrt{\langle (p - \langle p \rangle)^2 \rangle} = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$

$$\langle x \rangle = \text{Expectations value of } x$$

$$= \langle \psi | x | \psi \rangle$$

$$= \int \psi(x)^* x \psi(x) dx$$

$$= \int_{-\infty}^{+\infty} x \psi^2(x) dx$$

$$= \int_{-\infty}^{+\infty} x \left(\frac{a}{2\pi}\right)^{1/4} e^{-ax^2} dx = 0$$

$$\langle x^2 \rangle = \langle \psi | x^2 | \psi \rangle$$

$$\therefore \Delta x = \langle x^2 \rangle$$

$$= \int_{-\infty}^{\infty} \psi^* x^2 \psi(x) dx$$

$$\Delta p =$$

$$= \int_{-\infty}^{+\infty} x^2 \left(\frac{a}{2\pi} \right)^{1/4} e^{-ax^2} dx$$

$$= \left(\frac{a}{2\pi} \right)^{1/2} \int_0^{\infty} x^2 e^{-ax^2} dx$$

$$\int_0^{\infty} x^2 e^{-\frac{x^2}{a^2}} dx = \sqrt{\pi} \frac{2!}{1!} \left(\frac{a}{2} \right)^3$$

$$= 2 \left(\frac{a}{2\pi} \right)^{1/2} \int_0^{\infty} x^2 e^{-\frac{x^2}{b^2}} dx, \text{ where } b^2 = \frac{1}{a} \text{ so that } \langle x^2 \rangle = 2 \left(\frac{a}{2\pi} \right)^{1/2} \cdot \sqrt{\pi} 2 \left(\frac{1}{2\sqrt{a}} \right)^3$$

Quantum Structures

— simplest structure is

"no structure" i.e. $V(x)=0$, also called
free particle potential

— TISE

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \psi(x) = E \psi(x)$$

— For free particle, $-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} = E \psi$

Solutions

(i) Particular

$$\text{Let } k^2 = \frac{2mE}{\hbar^2}$$

$$\frac{d^2 \psi}{dx^2} = -k^2 \psi$$

$$\psi(x) = e^{\pm i k x} \quad \text{with } E = \frac{\hbar^2 k^2}{2m}$$

$P_{1,2}$

Particular solutions cannot be stationary states

(ii) General solution

Wavepacket

— Linear combination of states with different wavevectors (k).

Beats

Let us consider wavepacket of two components k_1, k_2 with equal weights

so that $\psi_{wp}(x) = e^{+jk_1x} + e^{+jk_2x}$

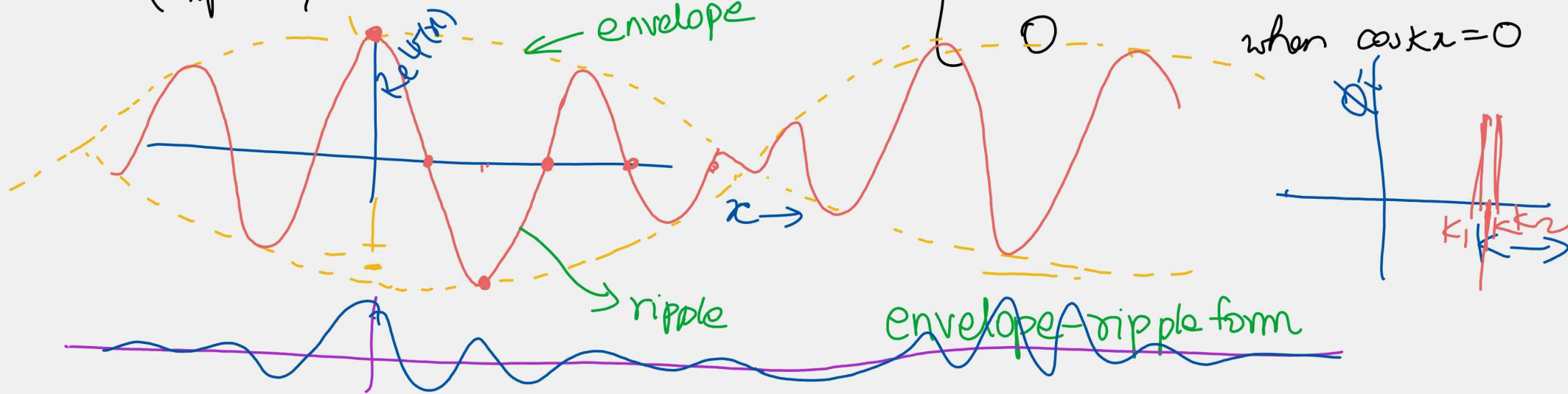
$$\text{Re}(\psi_{wp}(x)) = \cos k_1x + \cos k_2x$$

$$= 2 \cos\left(\frac{k_1+k_2}{2}x\right) \cos\left(\frac{k_1-k_2}{2}x\right)$$

Special case of "beats" wavepacket.

$k_1 \approx k_2 = k$, so that $k_1 - k_2 = \Delta k \ll k$

$$\text{Re}(\Psi_{\text{wp}}(x)) = 2 \cos kx \cos \frac{\Delta k}{2} x = \begin{cases} 2 \cos \frac{\Delta k}{2} x & \text{when } \cos kx = 1 \\ -2 \cos \frac{\Delta k}{2} x & \text{when } \cos kx = -1 \\ 0 & \text{when } \cos kx = 0 \end{cases}$$



If we increase the components



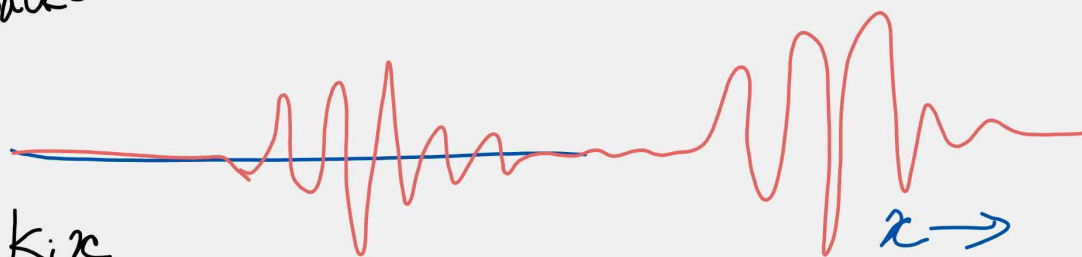
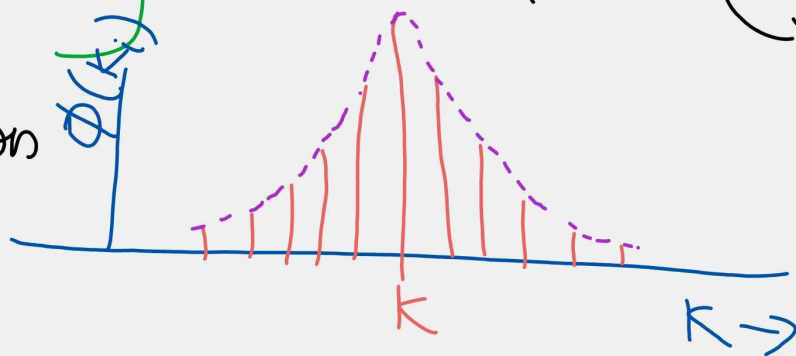
→ 3 component wavepacket

Recall

$$|\psi\rangle = \sum_{i=0}^{d-1} |i\rangle$$

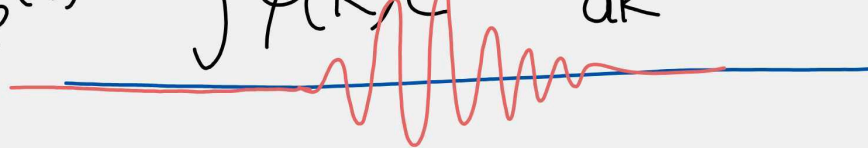
$$\psi_{wp}(x) = \sum_{i=1}^3 \phi(k_i) e^{+j k_i x}$$

By induction



∴ For the infinite component wavepacket

$$\psi_{wp}(x) = \int \phi(k) e^{+j k x} dk$$



Wavepacket $\rightarrow$ Components

Aim: Given a wavepacket, we should find the component (k) and its weight ($\phi(k)$)

Solution

Using Plancherel's theorem,

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi(x) e^{-jkx} dx$$

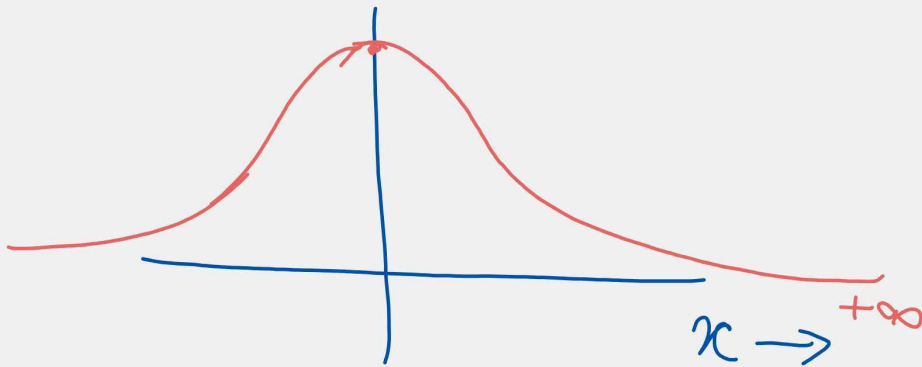
we have

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{+jkx} dk$$

Gaussian wavepacket

- Gauss
- Special case of wavepacket where the wavepacket has "Gaussian" form / inverted bell / normal distribution

$$\psi(x) = A e^{-ax^2}$$



Question : What are the components^(k) and the corresponding weights_{in} a Gaussian wavepacket?
($\phi(k)$)

$$\psi(x) = A e^{-ax^2}$$

If $\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{+jkx} dk$, then

$\phi(k)$, by Plancherel's theorem, is given by

$$\begin{aligned} \phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \psi(x) e^{-jkx} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} A e^{-ax^2} e^{-jkx} dx \end{aligned}$$

$$\phi(k) = \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-(ax^2 + jkx)} dx$$

Exponent of integrand is

$$\begin{aligned} &-(ax^2 + jkx) \\ &= -a\left(x^2 + j\frac{kx}{a}\right) \\ &= -a\left[\left(x + \frac{jk}{2a}\right)^2 - \frac{j^2 k^2}{4a^2}\right] \end{aligned}$$

$$= -a\left(x + j\frac{k}{2a}\right)^2 - \frac{k^2}{4a}$$

$$\therefore \phi(k) = \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-a\left(x + j\frac{k}{2a}\right)^2} e^{-\frac{k^2}{4a}} dx$$

$$\phi(k) = \frac{A}{\sqrt{2\pi}} e^{\frac{k^2}{4a}} \int_{-\infty}^{+\infty} e^{-a(x + \frac{jk}{2a})^2} dx$$

Gaussian integrals:

$$\int_0^{\infty} x^{2n} e^{-\frac{x^2}{b^2}} dx = \sqrt{\pi} \frac{(2n)!}{n!} \left(\frac{b}{2}\right)^{2n+1}$$

Using Gaussian integral for $n=0$,

$$\int_0^{\infty} e^{-\frac{x^2}{b^2}} dx = \sqrt{\pi} \frac{0!}{0!} \left(\frac{b}{2}\right) = \sqrt{\pi} \frac{b}{2}$$

$$\phi(k) = \frac{A}{\sqrt{2\pi}} e^{\frac{k^2}{4a}} \int_{-\infty}^{+\infty} e^{-ax^2} dx$$

After employing change of integration variable $x \rightarrow x + \frac{jk}{2a}$

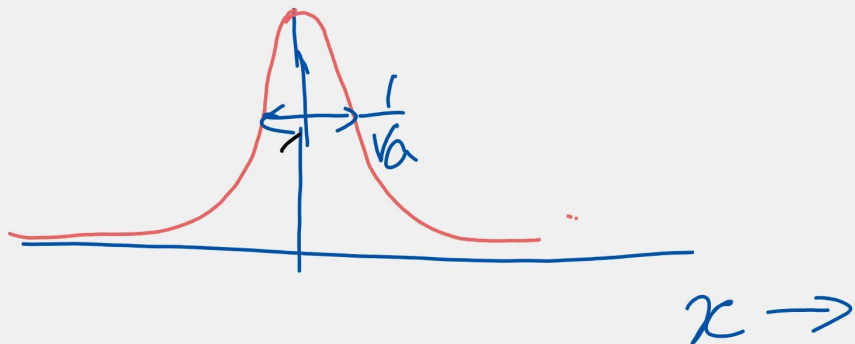
$$\begin{aligned} \phi(k) &= \frac{2A}{\sqrt{2\pi}} e^{\frac{k^2}{4a}} \int_0^{\infty} e^{-ax^2} dx \\ &= \frac{2A}{\sqrt{2\pi}} e^{\frac{k^2}{4a}} \int_0^{\infty} e^{-\frac{x^2}{b^2}} dx, \end{aligned}$$

$$\text{where } b^2 = \frac{1}{a}$$

$$= \frac{2A}{\sqrt{2\pi}} e^{\frac{k^2}{4a}} \sqrt{\pi} \frac{b}{2}$$

$$\phi(k) = \frac{A}{\sqrt{2a}} e^{\frac{k^2}{4a}}$$

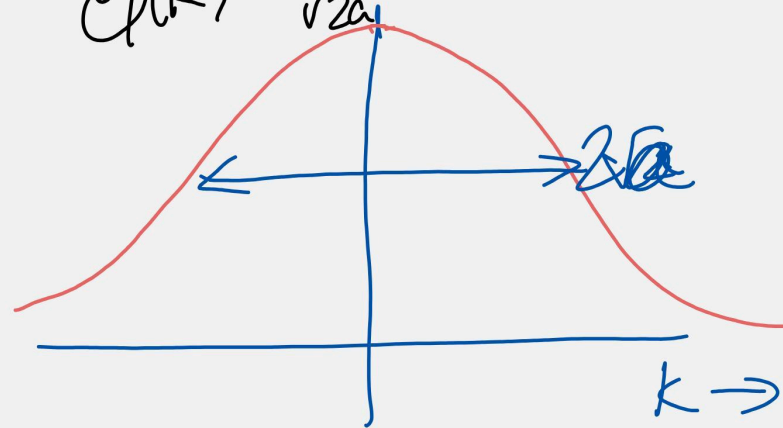
$$\psi(x) = A e^{-ax^2}$$



@ $x = \frac{1}{\sqrt{a}}$, $\psi(x) = A e^{-1} = \frac{A}{e}$

$$\Delta x$$

$$\phi(k) = \frac{A}{\sqrt{2a}} e^{-\frac{k^2}{4a}}$$



$$\Delta p$$

$$p = \hbar k$$

$$\frac{\hbar}{\lambda} \Rightarrow \frac{\hbar k}{2\pi}$$

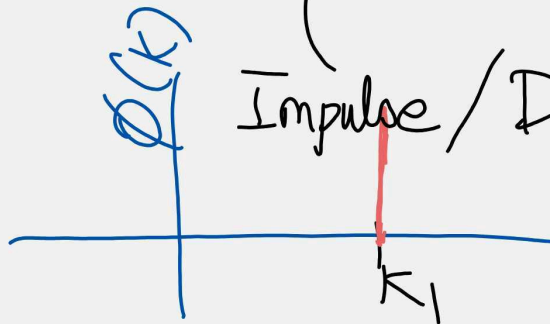
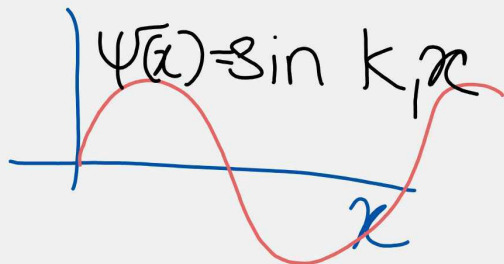
Gaussian wavepacket

$$\psi(x) = A e^{-ax^2}$$

is decomposition of e^{+jkx}

with weight $\phi(k)$ given by

$$\phi(k) = \frac{A}{\sqrt{2a}} e^{-\frac{k^2}{4a}}$$



$$\delta_{R.D.}(t) = T_0 \delta(t - t_0)$$

Impulse / Dirac delta function $\delta(x - x_0)$

Finite dimension
Kronecker delta function δ_{ij}

Infinite dimension

Normalized Gaussian wavepacket

Given $|\psi\rangle$, norm of $|\psi\rangle$ is

$$\langle\psi|\psi\rangle = 1 \quad // \text{Wavefunction postulate}$$

$$\Leftrightarrow \langle\psi|\tilde{I}|\psi\rangle = 1 \quad // \text{Completeness relation (Finite)} \quad \tilde{I} = \sum |i\rangle\langle i|$$

$$\Leftrightarrow \langle\psi|\int_{-\infty}^{+\infty} |x\rangle\langle x| dx |\psi\rangle = 1$$

$$\text{(Infinite)} \quad \tilde{I} = \int_{-\infty}^{+\infty} |x\rangle\langle x| dx$$

$$\Leftrightarrow \int_{-\infty}^{+\infty} \langle\psi|x\rangle \langle x|\psi\rangle dx = 1$$

$$// \langle x|\psi\rangle =: \psi(x)$$

$$\Leftrightarrow \int_{-\infty}^{+\infty} \psi^*(x) \psi(x) dx = 1$$

$$// \langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle^*$$

So $\psi(x) = Ae^{-ax^2}$

so $\langle \psi | \psi \rangle = \int_{-\infty}^{+\infty} A^* e^{-ax^2} A e^{-ax^2} dx = 1$

$\Leftrightarrow |A|^2 \int_{-\infty}^{+\infty} e^{-2ax^2} dx = 1$

$\Leftrightarrow 2|A|^2 \int_0^{\infty} e^{-2ax^2} dx = 1$

$\Leftrightarrow 2|A|^2 \int_0^{\infty} e^{-\frac{x^2}{b^2}} dx = 1$, where $b^2 = \frac{1}{2a}$

$\Leftrightarrow 2|A|^2 \frac{\sqrt{\pi}}{2} \frac{1}{\sqrt{2a}} = 1$

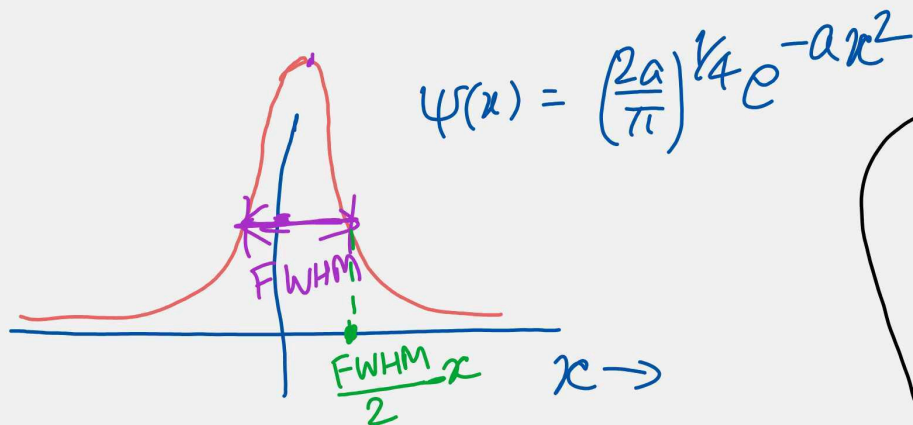
$\int_0^{\infty} e^{-\frac{x^2}{b^2}} dx = \frac{\sqrt{\pi}}{2} b$

$\Leftrightarrow A^2 = \sqrt{\frac{2a}{\pi}}$ // W.L.G. $A \in \mathbb{R}$

$\Leftrightarrow A = \left(\frac{2a}{\pi}\right)^{1/4}$

∴ normalised $\psi(x) = \left(\frac{2a}{\pi}\right)^{1/4} e^{-ax^2}$

and corresponding
k space spectrum
is $\phi(k) = \left(\frac{1}{2\pi a}\right)^{1/4} e^{-\frac{k^2}{4a}}$



$$\psi(x) = \left(\frac{2a}{\pi}\right)^{1/4} e^{-ax^2}$$

Full Width @ Half Maximum (FWHM)

$\psi(x)$ is maximum at $\bar{x} = 0$ and $\psi(0) = \left(\frac{2a}{\pi}\right)^{1/4}$

$$\text{So } \psi\left(\bar{x} + \frac{\text{FWHM}_x}{2}\right) = \frac{1}{2} \left(\frac{2a}{\pi}\right)^{1/4}$$

$$\Leftrightarrow \psi\left(\frac{\text{FWHM}_x}{2}\right) = \frac{1}{2} \left(\frac{2a}{\pi}\right)^{1/4}$$

$$\Leftrightarrow \left(\frac{2a}{\pi}\right)^{1/4} e^{-a\left(\frac{\text{FWHM}_x}{2}\right)^2} = \frac{1}{2} \left(\frac{2a}{\pi}\right)^{1/4}$$

$$e^{-a\left(\frac{\text{FWHM}_x}{2}\right)^2} = \frac{1}{2}$$

$$\Leftrightarrow -a\left(\frac{\text{FWHM}_x}{2}\right)^2 = \ln \frac{1}{2} = -\ln 2$$

$$\Leftrightarrow \frac{\text{FWHM}_x}{2} = \sqrt{\frac{\ln 2}{a}}$$

$$\Leftrightarrow \text{FWHM}_x = \frac{2\sqrt{\ln 2}}{\sqrt{a}}$$

i.e. $\text{FWHM}_x \propto \frac{1}{\sqrt{a}}$

Similarly $\phi(k) = \left(\frac{1}{2\pi a}\right)^{1/4} e^{-\frac{k^2}{4a}}$,

so $\text{FWHM}_k = 4\sqrt{\ln 2} \cdot \sqrt{a} \propto \sqrt{a}$

Heisenberg's uncertainty principle for Gaussian wavepacket $\left[\psi(x) = \left(\frac{2a}{\pi}\right)^{1/4} e^{-ax^2}\right]$

— $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$

— Δx = uncertainty in position observable

= (Root of (Mean of (Square of (Deviations of x 's from mean of x 's.

$$= \sqrt{\langle (x - \langle x \rangle)^2 \rangle}$$

So $\langle x \rangle = \langle \psi | \hat{x} | \psi \rangle$

$$= \int_{-\infty}^{+\infty} \psi^*(x) x \psi(x) dx$$

$$= \int_{-\infty}^{+\infty} x \left(\frac{2a}{\pi}\right)^{1/2} e^{-2ax^2} dx = 0$$

Gaussian integral for $n=1$

$$\int_0^{\infty} x^2 e^{-\frac{x^2}{b^2}} dx = \sqrt{\pi} \frac{2!}{1!} \left(\frac{b}{2}\right)^3 = \frac{\sqrt{\pi} b^3}{4}$$

$$\therefore \Delta x = \frac{1}{2\sqrt{a}}$$

$$\begin{aligned} \Rightarrow \Delta x &= \sqrt{\langle x^2 \rangle} \\ \langle x^2 \rangle &= \langle \psi | x^2 | \psi \rangle = \int_{-\infty}^{+\infty} \psi^*(x) x^2 \psi(x) dx \\ &= \int_{-\infty}^{+\infty} x^2 \left(\frac{2a}{\pi}\right)^{1/2} e^{-2ax^2} dx \\ &= 2 \left(\frac{2a}{\pi}\right)^{1/2} \int_0^{\infty} x^2 e^{-2ax^2} dx \\ &= 2 \left(\frac{2a}{\pi}\right)^{1/2} \frac{\sqrt{\pi}}{4} \left(\frac{1}{2a}\right)^{3/2} = \frac{2\sqrt{2}}{4} \frac{1}{2\sqrt{2}} \frac{1}{a} = \frac{1}{4a} \end{aligned}$$

$$\cancel{b^2 = \frac{1}{2a}}$$

— Δp — uncertainty in momentum observable

$$\psi(x) = \left(\frac{2a}{\pi}\right)^{1/4} e^{-ax^2}$$

$$- \Delta p = \sqrt{\langle (p - \langle p \rangle)^2 \rangle}$$

$$\text{So } \langle p \rangle = \langle \psi | \hat{p} | \psi \rangle$$

In coordinate basis, $\hat{p} \rightarrow \frac{\hbar}{i} \frac{d}{dx}$, so that

$$\langle p \rangle = \int_{-\infty}^{+\infty} \psi^*(x) \frac{\hbar}{i} \frac{d}{dx} \psi(x) dx$$

$$= \left(\frac{2a}{\pi}\right)^{1/2} \frac{\hbar}{i} \int_{-\infty}^{+\infty} e^{-ax^2} \frac{d}{dx} e^{-ax^2} dx$$

$$= \left(\frac{2a}{\pi}\right)^{1/2} \frac{\hbar}{i} \int_{-\infty}^{+\infty} e^{-ax^2} (-2ax e^{-ax^2}) dx = (\dots) \int_{-\infty}^{+\infty} x e^{-2ax^2} dx = 0$$

$$\text{So } \Delta p = \sqrt{\langle p^2 \rangle}$$

$$\text{Now, } \langle p^2 \rangle = \langle \psi | \hat{p}^2 | \psi \rangle$$

In coordinate basis,

$$\hat{p}^2 = \hat{p} \cdot \hat{p} = \frac{\hbar}{i} \frac{d}{dx} \left(\frac{\hbar}{i} \frac{d}{dx} \right) = \frac{\hbar^2}{i^2} \frac{d^2}{dx^2}$$

$$\hat{p}^2 = -\hbar^2 \frac{d^2}{dx^2}$$

$$\therefore \langle p^2 \rangle = \langle \psi | -\hbar^2 \frac{d^2}{dx^2} | \psi \rangle$$

$$= \int_{-\infty}^{+\infty} \psi^*(x) -\hbar^2 \frac{d^2}{dx^2} \psi(x) dx$$

$$= -\hbar^2 \left(\frac{2a}{\pi} \right)^{1/2} \int_{-\infty}^{+\infty} e^{-ax^2} \frac{d^2}{dx^2} e^{-ax^2} dx$$

$$= -\hbar^2 \left(\frac{2a}{\pi} \right)^{1/2} 2a \int_{-\infty}^{+\infty} e^{-ax^2} (2ax^2 - 1) e^{-ax^2} dx$$

$$= -\frac{\hbar^2 (2a)^{3/2}}{\pi^{1/2}} \left[\int_{-\infty}^{+\infty} 2ax^2 e^{-2ax^2} dx - \int_{-\infty}^{+\infty} e^{-2ax^2} dx \right]$$

$$\begin{aligned} &= -\frac{\hbar^2 (2a)^{3/2}}{\sqrt{\pi}} \left[2a \frac{\sqrt{\pi}}{4} \left(\frac{1}{2a} \right)^{3/2} - \frac{\sqrt{\pi}}{\sqrt{2a}} \right] \\ &= \frac{\hbar^2 (2a)^{3/2}}{\sqrt{\pi}} \left[\frac{\sqrt{\pi}}{4\sqrt{2a}} - \frac{\sqrt{\pi}}{\sqrt{2a}} \right] \\ &= \hbar^2 (2a) \frac{3}{4} \end{aligned}$$

$$\therefore \langle p^2 \rangle = \frac{3\hbar^2 \cdot 2a}{4} = \frac{3\hbar^2 a}{2}$$

$$\therefore \Delta p = \sqrt{\frac{3}{2}} \hbar \sqrt{a}$$

$$\begin{aligned}
 \text{So } \Delta x \cdot \Delta p &= \frac{1}{2\sqrt{a}} \cdot \sqrt{\frac{3}{2}} \hbar \sqrt{a} \\
 &= \frac{\hbar}{2} \sqrt{\frac{3}{2}} \\
 &\geq \frac{\hbar}{2}
 \end{aligned}$$



$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

vs

